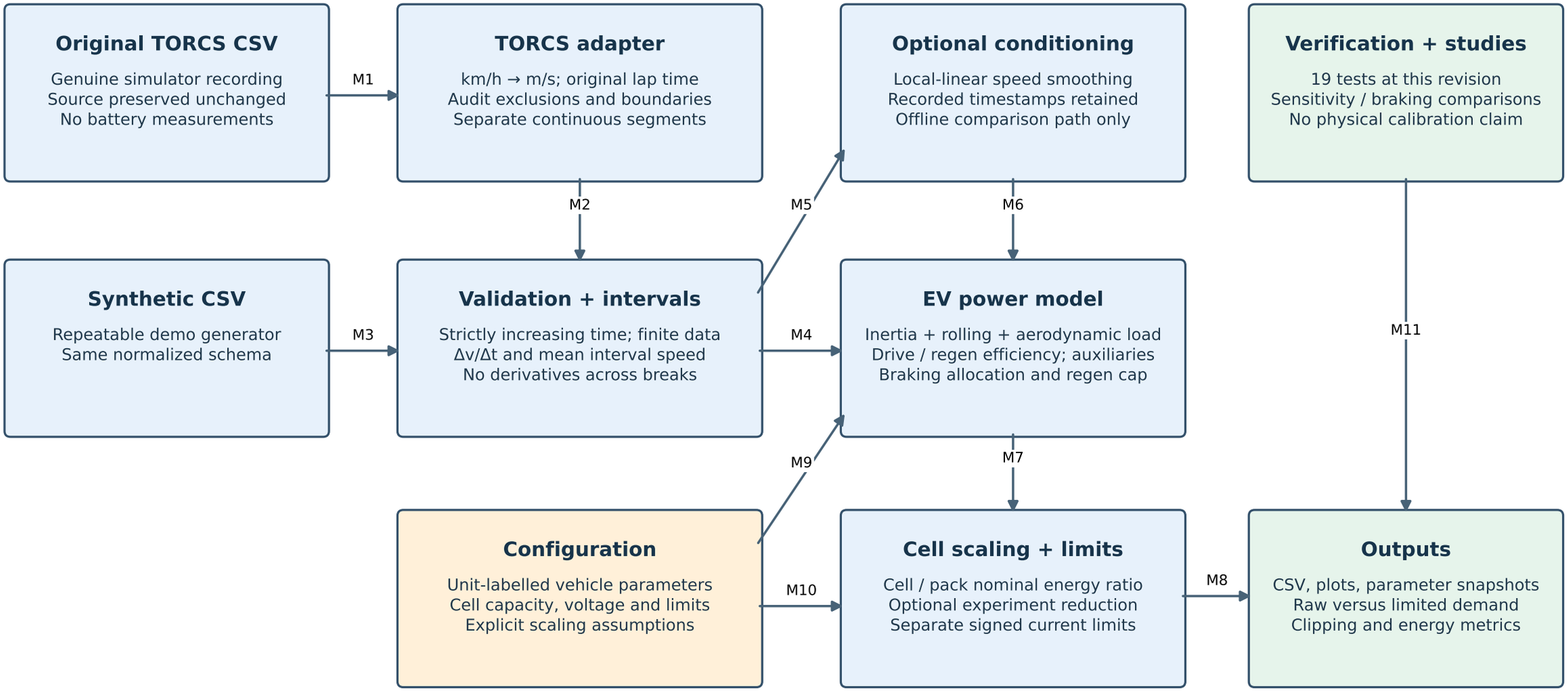
APS380 • Milestone 1: working offline software

Solid arrows = implemented files/functions. All currents are saved requests; no hardware or live simulator control.

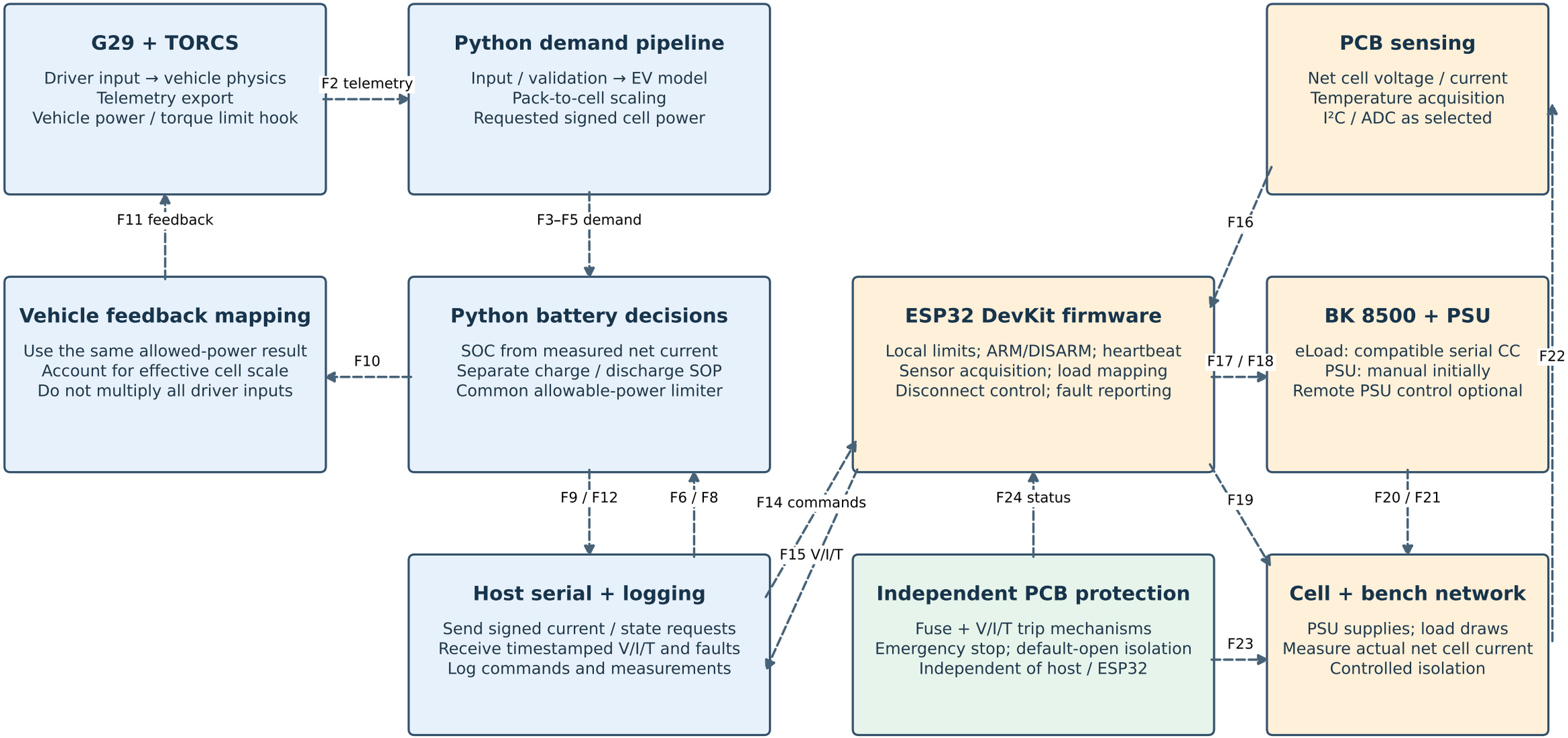


Positive current = discharge. Negative current = charge. First sample has zero integration duration.

Every connection and component is explained in SOFTWARE_ARCHITECTURE.md. Figure revision: October 8, 2026.

APS380 • intended closed-loop software architecture

Dashed arrows = planned integration. This overview groups internal connections; exact F1–F24 links are in the companion document.



F1 is G29 input inside the simulator block. F7 is SOP → limiter inside battery decisions. F13 is measurement logging.

Physical branch / isolation arrows provide context, not a wiring schematic. PSU bias must be verified, never assumed.

Python owns high-level decisions; ESP32 owns local execution; PCB protection must work independently. Revision: October 8, 2026.